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When Gender Arrives: Predictive Language Processing in the Third Year

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ABSTRACT

Predictive language comprehension allows children to use prior context to anticipate upcoming information before it becomes available in speech. Although young children use sentence meaning to guide real-time comprehension, less is known about when they generate more specific expectations about upcoming words without direct cues. Across two visual-world eye-tracking experiments, we used a grammatical contrast in Spanish as a test case to examine whether children aged 24, 30, and 36 months could use sentence context to guide attention before the expected word was heard. Highly predictable sentence contexts were followed by a silent interval before the final noun, allowing us to distinguish anticipatory looking from later word recognition. In Experiment 1, visual displays included a gender-matched object that shared grammatical gender with the expected noun and also carried transparent surface-form cues. In Experiment 2, those cues were removed, providing a stricter test of whether children could use internally generated expectations before word onset. Results showed anticipatory looks to the expected target across age groups, indicating context-based prediction. However, looks to the gender-matched object followed a more selective developmental pattern. With surface cues, the Gender-matched > Unrelated contrast was reliable at 30 and 36 months and larger at 36 than at 24 months. Without surface cues, it was reliable only at 36 months. These findings suggest that children's predictions develop from broad expectations about upcoming words toward more specific, linguistically structured expectations between 24 and 36 months. More generally, early predictive processing becomes increasingly precise as language representations become more stable and accessible.

Introduction

Prediction refers to the capacity of the cognitive system to generate expectations about upcoming events before they occur, relying on internal models shaped by prior experience to guide perception, action, and decision-making (Clark, 2013; de Lange, Heilbron, & Kok, 2018; Summerfield & de Lange, 2014). This anticipatory capacity operates across domains, from motor control—where forward models support online movement planning—to perception and higher-level cognition, allowing organisms to reduce processing costs and respond efficiently in real time (Körding & Wolpert, 2006; Wolpert & Flanagan, 2001). In language comprehension, prediction

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manifests as the pre-activation of upcoming linguistic representations at multiple levels, including semantic, morphosyntactic, and phonological features, before these are confirmed by the incoming speech signal (Kray et al., 2024; Ryskin & Nieuwland, 2023; Pickering & Gambi, 2018; Huettig, 2015; Kuperberg & Jaeger, 2016).

From a developmental perspective, prediction offers a window into how children integrate prior knowledge with unfolding input during real-time comprehension. A growing body of work shows that even in the second year of life, children anticipate upcoming referents based on semantic constraints conveyed by verbs and discourse context, as indexed by anticipatory eye movements in the visual-world paradigm (Borovsky et al., 2012; Mani & Huettig, 2012; Mani, Durrant, & Floccia, 2012). However, these early predictions appear to be relatively coarse-grained and primarily semantic, becoming increasingly fine-grained as children acquire more detailed lexical and grammatical knowledge (Gambi et al., 2018; Huettig, 2015). One key open question in developmental psycholinguistics concerns when children begin to anticipate morphosyntactic information and under what conditions such prediction becomes available during online processing.

Grammatical gender provides a particularly informative test case for studying the development of morphosyntactic prediction. Gender is a lexical–syntactic feature associated with nouns that is arbitrary, language-specific, and acquired gradually during development. In languages such as Spanish, gender agreement is pervasive and highly reliable, yet gender marking varies in transparency across lexical items, creating a natural contrast between regular and irregular forms. Adult research has shown that grammatical gender can be anticipated before noun onset when the discourse strongly constrains a single continuation. For example, electrophysiological studies with Spanish speakers demonstrate that gender expectations established by context or by pre-nominal elements elicit robust N400 and P600 effects when violated, indicating that gender features are pre-activated during comprehension rather than processed only after lexical access (Wicha, Moreno, & Kutas, 2003; Wicha et al., 2004; Wicha, Bates, Moreno, & Kutas, 2003). Similar anticipatory effects have been reported in other languages with gender agreement, including Polish (Szewczyk & Schriefers, 2013) and Italian (Ito et al., 2020), suggesting that morphosyntactic prediction is a general property of the adult language system when contextual constraints are sufficiently strong.

Developmental evidence, however, paints a more nuanced picture. Children's ability to exploit gender cues during online comprehension depends on both the transparency of the marking and the child's grammatical proficiency. In Spanish-learning children aged 34–42 months, Lew-Williams and Fernald (2007) showed that gender-marked articles (el "the" masculine, la "the" feminine) facilitate anticipatory looks toward the target object before the noun was fully available, with the magnitude of this effect correlating with individual differences in grammatical development. Similarly, Mornati et al. (2023) found that Italian-learning children aged 12 and 20 months used gender information carried by determiners during online comprehension: 12-month-olds identified the target only when the determiner was informative, that is, when the two displayed nouns differed in gender, whereas 20-month-olds recognized the target when it was named and were faster in the informative, different-gender condition. Other languages show a gradual developmental trajectory: in Russian, weak gender-based anticipatory effects emerge around age three and approach adult-like processing by age five (Aumeistere et al., 2022), while in French and Dutch reliable anticipatory use of gender appears only once children have established robust gender representations in production (Boloh & Ibernon, 2010; Brouwer, Sprenger, & Unsworth, 2017).

Crucially, most developmental studies reporting anticipatory effects provide overt gender information in the speech signal, typically through gender-marked determiners or adjectives (e.g., el "the" masculine, la "the" feminine). These studies provide important evidence that infants and young children can use gender information carried by determiners to pre-activate gender-related features of the upcoming noun before the noun itself is heard. However, because gender information is overtly available in the pre-nominal speech signal, they leave open a different question: Whether children can generate gender-based expectations when no overt gender cue is available before noun onset.

Spanish is particularly well suited for addressing this question because its grammatical gender system distinguishes masculine and feminine nouns and requires agreement on gender-marked articles and adjectives. For example, articles such as *el* "the.MASC" and *la* "the.FEM", as well as many adjectives, agree in gender with the noun they modify. At the same time, gender is only probabilistically related to noun form: many masculine nouns end in –o and many feminine nouns end in –a, but there are exceptions and nouns with opaque or atypical endings. Critically, some Spanish possessive determiners, including *mi* "my" and *su* "his/her/their," do not encode the gender of the following noun. This property makes it possible to create highly constraining pre-nominal contexts that withhold overt gender information, a design that is difficult to implement in languages such as Italian, where gender agreement is overtly marked across pre-nominal elements.

A related challenge concerns the role of phonological transparency. In many languages, including Spanish, noun endings (e.g., –a vs. –o) provide probabilistic cues to gender, which children learn early and exploit during comprehension (Seigneuric et al., 2007; van Heugten & Johnson, 2011). Consequently, observed anticipatory effects could be driven by sensitivity to phonological regularities rather than by abstract morphosyntactic features at the lexical level. Attempts to remove both determiner gender and transparent noun endings are rare in developmental research.

Together, these considerations leave open when Spanish-learning children can generate gender-based expectations from sentence context alone, in the absence of overt pre-nominal gender marking, and whether such expectations depend on transparent surface-form cues. The current study was designed to address this question.

The current study

The current study examined how Spanish-learning children's ability to use sentence context predictively changes across the third year of life. Specifically, we asked whether children can use highly predictable sentence contexts to anticipate upcoming words before they become available in the speech signal, and whether these predictions become increasingly specific with age. To address this

question, we used a visual-world eye-tracking paradigm with a 2000-ms silent interval immediately before noun onset, allowing us to distinguish anticipatory looking from later word recognition.

Children heard sentences such as *La gallina puso su… huevo* "The hen laid its… egg" while viewing two pictures. The use of possessive frames ensured that no direct gender cue to the final noun was provided immediately before word onset. Each sentence appeared in one of three two-picture display conditions. In the Target condition, the expected target picture was present together with a semantically and phonologically unrelated object whose noun shared the grammatical gender of the expected target noun; for example, children saw *huevo* "egg" and *lavabo* "sink," both masculine. In the Gender-matched condition, the expected target picture was absent. Instead, children saw a semantically and phonologically unrelated object whose Spanish noun shared the grammatical gender of the expected target noun, together with the Opposite-gender distractor, whose noun had the grammatical gender opposite to that of the target; for example, for the expected noun *huevo* "egg," children saw *cuchillo* "knife," which is masculine like *huevo*, and *gorra* "cap," which is feminine. In the Unrelated condition, the expected target picture was also absent; children saw the Unrelated object and the Unrelated filler, both of which shared the grammatical gender of the expected target noun and were semantically and phonologically unrelated to the expected target and to the sentence context, such as *lavabo* "sink" and *burro* "donkey," both masculine. The Unrelated object served as the critical object for analysis in this condition, whereas the Unrelated filler served as the distractor. Thus, the two displayed objects differed in grammatical gender only in the Gender-matched condition. The critical question was whether children would look more toward the Gender-matched object in the Gender-matched condition than toward the critical Unrelated object in the Unrelated condition before hearing the expected word itself.

Across two experiments, we varied the availability of surface-form cues in the Gender-matched object. In Experiment 1, the Gender-matched object shared grammatical gender with the expected noun and carried transparent, prototypical gender-marking cues. In Experiment 2, those surface cues were removed from the Gender-matched object by using nouns without transparent –o/–a gender-ending cues, providing a stricter test of whether children could use internally generated expectations to guide attention before the target word was heard. This contrast allowed us to ask whether children's predictions were driven primarily by broad contextual expectations and surface-level regularities, or whether they reflected access to more specific lexical–morphosyntactic information.

We tested children at 24, 30, and 36 months to characterize age-related changes in predictive processing during a period of rapid language development. We expected children to show evidence of context-based anticipation before noun onset, indexed by increased looks to the expected target. More importantly, we predicted that looks to the gender-matched object would become stronger and more selective with age, particularly when surface cues were unavailable. Such a pattern would suggest that early linguistic prediction develops from broad expectations about upcoming words toward more precise, linguistically structured expectations by the third year.

Experiment 1: Gender Prediction With Transparent Gender-Ending Cues

Method

Participants

The experiment included three age groups: 54 children at 24 months ($M = 23.97$ months, $SD = 0.29$; 24 boys), 53 children at 30 months ($M = 30.01$ months, $SD = 0.26$; 29 boys), and 43 children at 36 months ($M = 35.99$ months, $SD = 0.28$; 21 boys). All participants were Mexican children acquiring Spanish as their only language, with normal hearing and vision and no parent-reported neurological or language impairments. Standardized developmental measures differed by age group because of the age ranges of

Table 1
Participant assessment descriptives by age group

Domain	Measure	24 months M (SD), range	30 months M (SD), range	36 months M (SD), range
Socioeconomic background	Monthly household income	MXN 12,519 (1,274), 10,592–14,731	MXN 12,535 (1,212), 10,562–14,747	MXN 12,504 (1,233), 10,553–14,652
CDI-II	Words produced	502.46 (49.95), 400–598	487.58 (28.53), 452–538	—
	Word combinations	6.28 (0.79), 5–7	5.26 (1.09), 4–7	—
	Syntactic complexity	12.19 (1.95), 9–16	26.62 (3.63), 21–32	—
WPPSI-III	Puzzles	—	16.51 (1.09), 15–18	21.26 (2.24), 18–25
	Block Design	—	21.55 (2.10), 19–25	24.95 (3.19), 20–29
	Information	—	15.26 (2.25), 12–18	17.40 (2.54), 14–21
	Receptive Vocabulary	—	21.21 (2.01), 18–24	23.00 (1.59), 20–25
	Naming	—	15.23 (1.09), 14–17	18.77 (2.01), 16–22
	Performance IQ	—	99.83 (4.14), 91–109	100.98 (6.79), 88–118
	Verbal IQ	—	101.13 (6.48), 88–112	100.35 (6.84), 88–115
	Full-scale IQ	—	100.49 (3.82), 94–109	100.70 (4.69), 88–112

Note. Values are M (SD), range, unless otherwise indicated. Monthly household income is reported in Mexican pesos. As a national benchmark, the 2024 average monthly household income in Mexico was MXN 25,955 according to the National Survey of Household Income and Expenditure (ENIGH; INEGI, 2025); therefore, the sample means reported here were below the national household average and should be interpreted as descriptive socioeconomic information. CDI-II = MacArthur–Bates Communicative Development Inventories. CDI-II words produced corresponds to the number of words children were reported to produce; word combinations indexes the number of words children combined in utterances; syntactic complexity indexes children's use of more complex syntactic structures. WPPSI-III = Wechsler Preschool and Primary Scale of Intelligence, Third Edition. Dashes indicate that the measure was not administered to that age group.

the instruments. The MacArthur–Bates Communicative Development Inventories (CDI-II; Jackson-Maldonado et al., 2003), standardized for Mexican Spanish, were used to characterize language development, whereas selected subtests from the Wechsler Preschool and Primary Scale of Intelligence, Third Edition (WPPSI-III; Wechsler, 2011), were used to characterize cognitive abilities in age-eligible children. The 24-month-old children completed only the CDI-II, 30-month-old children completed both the CDI-II and selected WPPSI-III subtests, and 36-month-old children completed only the WPPSI-III because the CDI-II is designed for children up to 30 months. Participant socioeconomic information and standardized assessment descriptives, including ranges, are summarized in Table 1.

A sensitivity analysis was performed to determine the minimum effect size detectable with the available sample (Lakens, 2022). The analysis focused on the critical anticipatory effect: the gender-matched object advantage (Gender-matched > Unrelated) during the prediction window (−2000 to 0 ms), which indexes morphosyntactic, gender-based prediction.

Sensitivity estimates were computed, assuming a two-tailed alpha level of .05 and a desired statistical power of .80, following standard conventions in psychological research (Cohen, 1988; Lakens, 2022). Based on the available sample sizes, the study was sufficiently powered to detect medium-sized effects, corresponding to Cohen's $d \approx 0.45$. This estimate aligns with conventional benchmarks for medium effects (Cohen, 1988) and is consistent with effect sizes typically reported in visual-world eye-tracking studies with young children (e.g., Mani & Huettig, 2012; Borovsky et al., 2012). The final sample sizes for the 24-month ($n = 54$) and 30-month ($n = 53$) groups met or exceeded this threshold, while the 36-month group ($n = 43$) approached it closely. These estimates take into account the variability and data loss characteristic of eye-tracking research with young children (Gredebäck et al., 2010). Therefore, the present design provides adequate sensitivity to detect developmentally relevant changes in anticipatory processing across age groups.

This research was carried out in accordance with the Declaration of Helsinki of the World Medical Association and its ethical principles for medical research involving human subjects. The study was non-invasive and posed no risk to the participants or their parents or guardians. Participation was voluntary, and informed consent was obtained from parents or caregivers. All collected data were stored securely and treated confidentially.

The research protocol was approved by the Ethics Committee of the Master's and Doctoral Program in Psychology at the National Autonomous University of Mexico (UNAM), as documented in official approval letter No. EP/PMDPSIC/0125/2025.

Data, code, and preregistration

The eye-tracking data and R analysis scripts used for Experiment 1 and Experiment 2 are openly available on the Open Science Framework (OSF): https://osf.io/xzgvk/overview?view_only=22d71a4173ed4048bef52b63d800cb45. These materials include the processed visual-fixation data and the code used to conduct the statistical analyses reported in the present manuscript. No part of the study design, hypotheses, or analyses reported in this manuscript was preregistered.

Experimental design

The temporal sequence of all trials was designed to disentangle anticipatory processes from processes associated with auditory integration (see Fig. 1). At the beginning of each trial, participants fixated on a central cross, after which, at −4500 ms, two images were presented simultaneously. Immediately thereafter, participants heard the sentence context, for example, *Ayer, la gallina puso su…* (Yesterday, the hen laid its…). The sentence context ended at −2000 ms, followed by a silent pause lasting until the onset of the spoken target noun at 0 ms. This interval, defined as the prediction window, provided a crucial period for examining whether participants used the sentence context to anticipate the upcoming noun or gender-related information associated with that noun in the absence of additional auditory input.

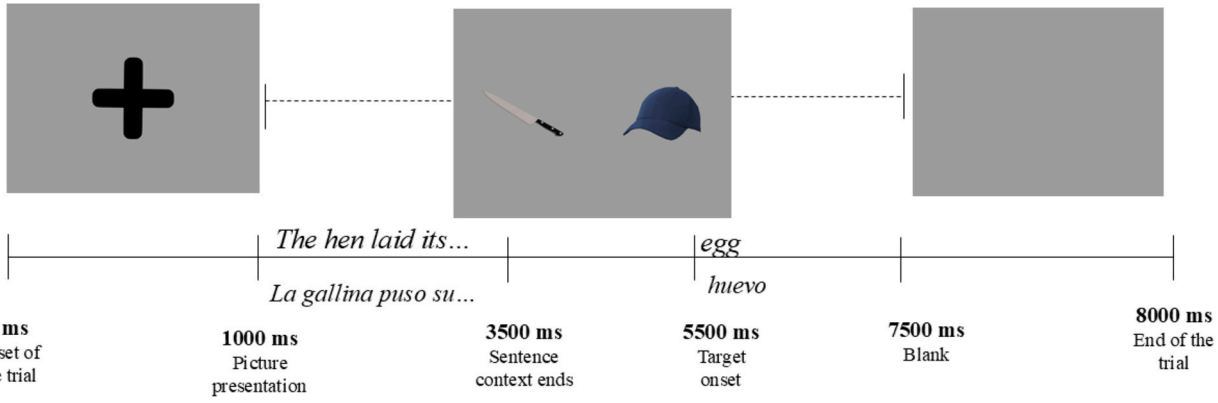


Fig. 1. Temporal sequence of each trial. In Spanish, *cuchillo* ("knife") is grammatically masculine, matching the gender of the expected noun *huevo* ("egg"), whereas *gorra* ("cap") is grammatically feminine.

At 0 ms, the spoken target noun was presented (e.g., "egg"), marking the beginning of the integration window, which extended from 0 to 2000 ms. During this interval, gaze patterns reflected lexical recognition and the bottom-up integration of the auditory input with the corresponding visual referent. Because the target picture was not present in all conditions, post-noun fixations were interpreted as reflecting recognition of the target when it was visually available, or integration of the spoken noun with the available alternatives when the target was absent. In this way, the experimental timeline allowed us to analyze anticipatory fixations during the silent pre-noun interval separately from fixations following noun onset.

The 21 sentence frames were simple, grammatically correct, and highly constraining. They were constructed around familiar nouns drawn from the MacArthur–Bates Communicative Development Inventories for Mexican Spanish. Sentence frames were designed to avoid overt pre-nominal gender marking of the target noun, most commonly by using possessive determiners such as *mi* "my" and *su* "his/her/their," which do not encode grammatical gender in Spanish. Thus, the design minimized overt gender information in the speech signal before the target noun was heard. Full stimulus lists and validation details are provided in the Supplementary Materials; the structure of the three display conditions is summarized below.

Each sentence appeared in one of three two-picture display conditions: Target, Gender-matched, or Unrelated. The condition label refers to the critical object analyzed in that display, whereas the other picture served as the distractor. The item labeled as the Unrelated object in the stimulus tables served as the non-target distractor in Target trials and as the critical object in Unrelated trials, whereas the Unrelated filler served as the distractor only in Unrelated trials. The conditions differed as shown in Table 2.

Experimental conditions

Thus, in the Target condition, the expected referent was visually present together with the Unrelated object, whose noun shared the grammatical gender of the expected target noun. In the Gender-matched condition, the expected target picture was absent, but the display included the Gender-matched object, that is, an unrelated object with the same grammatical gender as the expected target, together with the Opposite-gender distractor, whose noun had the other grammatical gender. In the Unrelated condition, the target picture was also absent, and the display contained the Unrelated object and the Unrelated filler, both of whose nouns shared the grammatical gender of the expected target. Thus, grammatical gender distinguished between the two displayed pictures only in the Gender-matched condition. The critical comparison for gender-based prediction was therefore looks to the Gender-matched object in the Gender-matched condition relative to looks to the critical Unrelated object in the Unrelated condition during the prediction window.

The 21 sentences were divided into three sets of seven items across the three experimental conditions: Target, Gender-matched, and Unrelated. To ensure full counterbalancing, we constructed six lists following a Latin-square design. Across lists, each sentence appeared equally often in each condition, such that an item presented in the Target condition in one list appeared in the Gender-matched and Unrelated conditions in other lists. The six lists also counterbalanced the left–right position of the critical and distractor images, so that critical objects appeared equally often on each side of the screen across lists.

Each participant was assigned to a single list and therefore saw each sentence only once, in one condition. As a result, participants were exposed to a total of 21 trials: 7 Target, 7 Gender-matched, and 7 Unrelated trials. Within each list, the left–right position of images was also counterbalanced, such that the critical object and the distractor appeared equally often on each side of the screen.

Experiment 1 examined morphosyntactic prediction when the Gender-matched objects carried transparent, prototypical gender-marking endings in Spanish, for example, -o, typically masculine, and -a, typically feminine.

This design allowed us to assess whether children anticipate grammatical gender based on contextual information in the absence of overt pre-nominal cues. Specifically, increased looks to the Gender-matched object relative to the Unrelated object before noun onset would indicate that children used the sentence context to pre-activate gender information associated with the expected noun.

Instruments

To gather general and health-related information about participants, parents or legal guardians completed a sociodemographic questionnaire. This instrument included items on household income, attendance at early intervention programs, preschool or daycare, exposure to a second or third language, and the child's overall health. These data were used to characterize the sample and to apply exclusion criteria. Any child whose parent reported hearing, neurological, or language problems, or regular exposure to another language, was excluded from analysis. Household income was used as a descriptive index of socioeconomic background and is reported in the Participants section.

In addition to the sociodemographic questionnaire, two standardized developmental assessments were administered to characterize children's language and cognitive profiles and to explore whether individual differences modulated anticipatory looking. Because these analyses were exploratory and involved different measures across age groups, they were interpreted cautiously and were not treated as primary tests of the study hypotheses.

Table 2

Condition	Pictures presented	Critical object	Example for Ayer, la gallina puso su… huevo "Yesterday, the hen laid its… egg"
Target	Target + Unrelated object	Target	huevo "egg" + lavabo "sink"
Gender-matched	Gender-matched object + Opposite-gender distractor	Gender-matched object	cuchillo "knife," masculine like huevo, + gorra "cap," feminine
Unrelated	Unrelated object + Unrelated filler	Unrelated object	lavabo "sink" + burro "donkey"

First, parents or guardians completed the *Inventario de Desarrollo de Habilidades Comunicativas MacArthur-Bates CDI-II* "MacArthur–Bates Communicative Development Inventories (CDI–II)" (Jackson-Maldonado et al., 2003) standardized for Mexican Spanish. The CDI-II provided parent-report measures of children's productive vocabulary, word combinations, and syntactic complexity. Specifically, we used the number of words produced, the number of words children combined to form utterances, and a syntactic-complexity score indexing children's use of more complex grammatical structures. It was administered only to the 24- and 30-month-old children, because the instrument is designed for infants aged 18–30 months and is not appropriate for 36-month-olds. This inventory was included as previous studies have demonstrated a relationship between productive vocabulary skills and linguistic prediction (Mani & Huettig, 2012).

The second instrument consisted of selected subtests from the *Wechsler Preschool and Primary Scale of Intelligence-III* (Wechsler, 2011), which is standardized for the Mexican population for children aged 2 years and 6 months to 7 years and 3 months. WPPSI-III was administered to the 30- and 36-month-old children. It was not administered to the 24-month-old group because these children were below the age range of the instrument. The selected subtests used were as follows:

Receptive Vocabulary: Participants viewed a set of four images and indicated the one that was named.

Information: Participants answered a question by choosing from four image options.

Puzzles: Participants assembled puzzle pieces presented in a standard layout into a figure within a time limit.

Block Design: Participants reproduced a model or image using plastic cubes within a time limit.

Naming: Participants named visually presented objects.

We assessed whether individual differences modulated anticipatory behavior, analyzing each age group separately because the assessment batteries differed across cohorts. For the 24-month group, predictors were CDI-II indices of words produced, word combinations, and syntactic complexity. For the 30-month group, the same CDI-II measures were included together with the WPPSI-III Full-scale IQ as an index of cognitive development. For the 36-month group, only the WPPSI-III Full-scale IQ was evaluated as the cognitive development index.

Procedure

The parent/guardian and child were welcomed in a reserved space with no foot traffic. The study was explained in age-appropriate language. Written informed consent was obtained from the guardian, and assent was obtained from the child when appropriate. A brief sociodemographic questionnaire was administered, and basic functional sensory criteria were verified.

Sessions were conducted in situ ensuring a quiet setting (no nearby activity) and stable ambient illumination throughout. Children were seated approximately 60–65 cm from a 1920×1080 px monitor. Eye movements and pupil size were recorded with a *Tobii TX-120* system using a 5-point calibration at the start of the session and verification prior to each block. Short breaks were offered as needed. Validity and exclusion criteria were homogeneous across experiments (invalid calibration or excessive tracking loss, or participant discomfort).

Linguistic and Visual Stimuli

The eye-tracking task included 21 highly constraining Spanish sentence frames ending in familiar nouns. Target nouns were selected from the MacArthur–Bates Communicative Development Inventories for Mexican Spanish, and all had a familiarity rating of at least 60% in WordBank (Frank et al., 2017). Sentences were simple, grammatically correct, and constructed to avoid overt pre-nominal gender marking, most commonly through possessive determiners such as mi "my" and su "his/her/their," which do not encode grammatical gender in Spanish.

In Experiment 1, the nouns corresponding to the Gender-matched objects had transparent, prototypical gender endings in Spanish, such as –o for masculine nouns and –a for feminine nouns.

Visual stimuli consisted of color photographs of familiar objects presented in two-picture displays on a gray background. Images were selected to be recognizable to young children and were edited to standardize size and background. Across the full stimulus set, all non-target objects were selected to be semantically and phonologically unrelated to the expected target noun and to the sentence context.

Full stimulus lists, counterbalancing information, phonological and semantic similarity checks, image-naming results, and visual-saliency analyses are reported in the Supplementary Materials.

Data Processing and Statistical Analysis

We analyzed looking behavior at two complementary levels. First, we used window-averaged mixed-effects models to test overall differences between conditions and age groups within the theoretically defined prediction and integration periods. These models constituted the primary inferential analyses and allowed us to account for variability across participants and items. Second, we used a cluster-based permutation analysis to describe the temporal distribution of the condition effects across the trial.

Data processing was conducted in R (version 4.3.2; R Core Team, 2023) using the eyetrackingR package (Dink & Ferguson, 2015). Fixation proportions were calculated for two areas of interest corresponding to the target and distractor images. Gaze samples were coded as track loss only when data from both eyes were invalid; samples with valid data from at least one eye were retained. Trials with more than 80% track loss during the 6,500-ms image-presentation period were excluded. No additional participant-level exclusion criterion based on a minimum number of valid trials was applied; however, all participants contributed at least one valid trial per

condition, with an average of 19.26 (SD = 3.77, range = 3 - 21) out of 21 valid trials.

In the 24-month-old group, participants contributed an average of 6.50 (SD = 0.66, range = 5 - 7) valid target trials, 6.62 (SD = 0.85, range = 4 - 7) valid Gender-matched trials, and 6.61 (SD = 0.83, range = 4 - 7) valid unrelated trials. In the 30-month-old group, the corresponding averages were 6.71 (SD = 0.60, range = 5 - 7) target trials, 6.90 (SD = 0.40, range = 5 - 7) Gender-matched trials, and 6.88 (SD = 0.57, range = 4 - 7) unrelated trials. In the 36-month-old group, children contributed an average of 5.62 (SD = 2.07, range = 1 - 7) target trials, 5.74 (SD = 2.05, range = 1 - 7) Gender-matched trials, and 5.74 (SD = 2.05, range = 1 - 7) unrelated trials.

Our main analysis consisted of two different linear mixed-effects models implemented using the *lmer* function from the *lme4* package (Bates et al., 2015). The first model measured anticipatory looks during the prediction window (−2000 to 0 ms), when a pause was introduced before target onset. The second model assessed bottom-up effects during the integration window (0 to 2000 ms), after the target was heard. The dependent variable was the average of empirical logits[1] calculated over these time windows (Ito & Knoeferle, 2023). We included two fixed factors using a dummy coding scheme: *Group*, with three levels (24, 30, and 36 months) and 24 months as the reference level; and *Condition*, with three levels (Unrelated, Gender-matched, Target) and Unrelated as the reference level. We specified the random effects structure to include the maximum number of terms justified by the experimental design (Barr, 2008). We therefore modeled Condition as a random slope by subject and included a random intercept for items. The formula for the model was as follows:

$$\mathrm{Elog}_{ij} \approx \beta_0 + \beta_1 \cdot \mathrm{Condition}_{ij} + \beta_2 \cdot \mathrm{Group}_{ij} + \beta_3 \cdot \left(\mathrm{Condition}_{ij} \times \mathrm{Group}_{ij}\right) + (S_{0i} + S_{1i} \cdot \mathrm{Condition}_{ij}) + I_{0j} + \epsilon_{ij}$$

Where:

β_0: The overall intercept.
β_1: The fixed effect coefficient for Condition effect.
β_2: The fixed effect coefficient for Group effect.
β_3: The coefficient for interaction between Condition and Group effects.
S_{0i}: The random intercept for Subject *i*.
S_{1i}: The random slope for Condition effect for Subject *i*.
I_{0j}: The random intercept for Item *j*.
ϵ_{ij}: The residual error term.

To characterize the age-specific pattern underlying the Condition × Group effects, estimated marginal means were obtained using the *emmeans* package. Reported follow-up contrasts were restricted to the theoretically critical Gender-matched versus Unrelated comparison within each age group and were Bonferroni-adjusted for multiple testing. These within-group contrasts were used to determine whether a Gender-matched-object advantage was detectable at each age; developmental differences between age groups were inferred only from the corresponding Condition × Group interaction terms.

As a complementary analysis, we conducted a cluster-based permutation analysis to characterize the temporal distribution of condition differences across the trial (Maris & Oostenveld, 2007). Unlike the mixed-effects models, this analysis was not intended to estimate an additional overall condition effect or to provide precise inferential estimates of effect onset or duration. Rather, it was used to identify whether significant clusters overlapped with the pre-noun prediction period, the post-noun integration period, or both. The onset and offset of significant clusters were not statistically compared across conditions or age groups and were therefore interpreted descriptively. The data were binned into 100 ms intervals to reduce autocorrelation (Mirman, 2014), and each bin was aggregated at the subject level. The analysis covered the time window from −4000 to 2000 ms, encompassing the baseline, the prediction window, and the integration window. Comparisons were limited to target vs. unrelated images and gender-matched vs. unrelated images.

Comparisons were performed using *t*-tests with a cluster-based statistic, defined as the sum of adjacent *t*-values exceeding the critical threshold corresponding to $\alpha = 0.05$. The null distribution was generated from 100,000 random permutations of condition labels (Pernet et al., 2015), conducted independently for the target and gender-matched contrasts. The p-value was calculated as the proportion of cluster statistics from the permutation distribution that exceeded the observed cluster statistic, using a significance threshold of $\alpha = 0.05$.

In addition, we explored whether individual differences modulated anticipatory effects. As different assessments were administered to each age group, this analysis was conducted separately for each group. For the 24-month-old group, we included CDI scores for total vocabulary, word combinations, and syntactic complexity. For the 30-month-old group, we included CDI scores as well as a measure of cognitive development. For the 36-month-old group, only cognitive development was evaluated. All assessment scores were centered and scaled to improve model convergence. Based on the descriptive time-course pattern from the cluster-based analysis, we used the average empirical logits over the −1000 to 0 ms window as the dependent variable. This data-informed interval was used only for the exploratory individual-difference analyses and was not interpreted as an inferential estimate of effect onset or duration.

Thus, for each group, we fitted a mixed-effects model similar to the main analysis as a reference model, including only the fixed

[1] The empirical logit was calculated as (log[(y + 0.5)/(n - y + 0.5)]), where (y) is the number of samples in which the participant looked at the area of interest and (n) is the total number of valid samples. The addition of 0.5 prevents undefined values when the observed proportion is 0 or 1. We used this transformation because raw proportions are bounded between 0 and 1 and typically exhibit nonconstant variance, particularly near the boundaries. Consequently, the empirical-logit transformation reduces these boundary and heteroscedasticity problems and provides a response variable that is more appropriate for parametric analysis.

factor Condition, while keeping the same random-effects structure. In a second step, we fitted a model that additionally included one of the evaluation scores and its interaction with Condition (e.g., Condition × Vocabulary). The two models were compared using a log-likelihood ratio test (χ^2). To control for multiple comparisons across the eight individual-difference models fitted within each experiment, we applied a Bonferroni correction, resulting in a corrected significance threshold of $\alpha = .00625$ (.05/8). Only models that significantly improved fit under this criterion were considered for follow-up interpretation.

Results

Linear mixed effect models

Fig. 2 illustrates the time course of the fixation patterns across the analysis window, and Table 3 presents the full statistical results of the linear mixed-effect models. In the prediction window, significant effects of Condition were observed for both the Target ($p < 0.001$) and Gender-matched ($p = 0.043$) conditions, indicating that the average empirical logit fixation values were significantly higher in these conditions compared to the Unrelated condition. The Gender-matched × 36 months interaction was also significant ($p < 0.001$), showing that the difference in fixations between the Gender-matched and Unrelated conditions was significantly larger at 36 months than at 24 months. In contrast, the Gender-matched × 30 months interaction was not significant ($p = 0.373$), indicating no reliable change in the gender-matched effect at 30 months relative to 24 months.

The Bonferroni-adjusted follow-up contrasts showed that in the 24-month group, there was no significant difference in fixations between the Unrelated and Gender-matched conditions ($\beta = -0.129$, $SE = 0.063$, $df = 137$, $t = -2.046$, $p = 0.128$). In contrast, in the 30-month group, fixations differed significantly between the Unrelated and Gender-matched conditions ($\beta = -0.209$, $SE = 0.062$, $df = 126$, $t = -3.360$, $p = 0.003$). Similarly, in the 36-month group, the contrast between Unrelated and Gender-matched conditions was significant ($\beta = -0.610$, $SE = 0.075$, $df = 147$, $t = -8.074$, $p < 0.001$).

In the integration window, significant effects of Condition were observed for both the Target ($p < 0.001$) and Gender-matched ($p = 0.006$) conditions, indicating that the average empirical logit of fixations was significantly higher in these conditions than in the

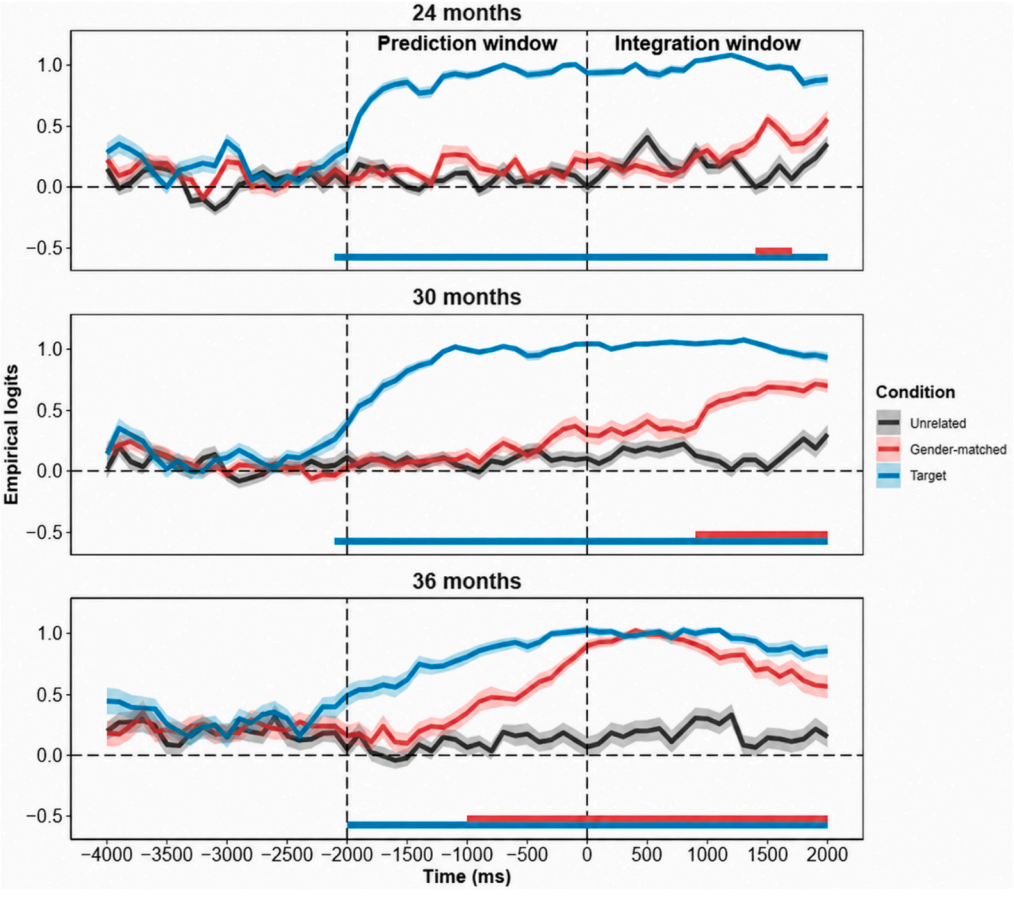


Fig. 2. Fixation Time Course for Experiment 1. Note. Lines represent the mean empirical logits, with shaded ribbons indicating ± 1 standard error. Vertical dashed lines mark the analysis windows, and horizontal dashed lines indicate chance level. Solid horizontal bars at the bottom denote significant clusters: red for the gender-matched-unrelated contrast, and blue for the target-unrelated contrast.

Table 3
Linear Mixed-Effects Models for Experiment 1

Fixed effects	Prediction window					Integration Window				
	β	*SE*	*df*	*t*	*p*	*β*	*SE*	*df*	*t*	*p*
Intercept	-0.067	0.045	263.582	-1.466	0.143	0.002	0.035	435.6	0.064	0.948
Gender-matched	**0.128**	**0.063**	**413.213**	**2.027**	**0.043**	**0.16**	**0.058**	**240.9**	**2.755**	**0.006**
Target	**0.991**	**0.058**	**2695.955**	**16.823**	**< 0.001**	**0.926**	**0.048**	**2663**	**19.195**	**< 0.001**
30 months	-0.031	0.058	2694.541	-0.539	0.589	-0.039	0.047	2722	-0.834	0.404
36 months	-0.032	0.065	2695.823	-0.493	0.622	-0.049	0.053	2723	-0.936	0.349
Gender-matched : 30 months	0.079	0.089	398.67	0.893	0.372	**0.282**	**0.082**	**235.3**	**3.441**	**< 0.001**
Target : 30 months	0.076	0.082	2691.766	0.929	0.352	0.094	0.067	2656	1.389	0.164
Gender-matched : 36 months	**0.481**	**0.099**	**431.404**	**4.849**	**< 0.001**	**0.673**	**0.09**	**254.8**	**7.422**	**< 0.001**
Target : 36 months	0.049	0.092	2693.577	0.539	0.589	0.054	0.075	2663	0.716	0.473

Note. Bold values indicate statistical significance.

Unrelated condition. The Gender-matched × 30 months ($p < 0.001$) and Gender-matched × 36 months ($p < 0.001$) interactions were also significant, showing that the difference in fixations between the Gender-matched and Unrelated conditions was significantly larger at both 30 and 36 months compared to 24 months. In contrast, the Target × 30 months ($p = 0.404$) and Target × 36 months ($p = 0.473$) interactions were not significant, indicating that the target effect did not significantly differ across age groups.

Bonferroni-adjusted follow-up contrasts revealed that in the 24-month group, there was a significant difference in fixations between the Unrelated and Gender-matched conditions ($\beta = -0.161$, $SE = 0.061$, $df = 171$, $t = -2.628$, $p = 0.028$). In the 30-month group, fixations differed significantly between the Unrelated and Gender-matched conditions ($\beta = -0.443$, $SE = 0.060$, $df = 164$, $t = -7.291$, $p < 0.001$). In the 36-month group, there were significant differences between Unrelated and Gender-matched conditions ($\beta = -0.833$, $SE = 0.0727$, $df = 185$, $t = -11.463$, $p < 0.0001$).

Cluster-based permutation analysis

In the target vs. unrelated comparison, clusters extended from −2100 to 2000 ms in the 24-month-old group ($t_{cluster} = 468.26$, $t_{max} = 17.31$, $p < .001$), from −2100 to 2000 ms in the 30-month-old group ($t_{cluster} = 522.70$, $t_{max} = 18.28$, $p < .001$), and from −2000 to 2000 ms in the 36-month-old group ($t_{cluster} = 340.31$, $t_{max} = 11.58$, $p < .001$). In contrast, for the gender-matched vs. unrelated comparison, a significant cluster was observed from 1400 to 1700 ms in the 24-month-old group ($t_{cluster} = 12.61$, $t_{max} = 3.36$, $p = .003$), from 900 to 2000 ms in the 30-month-old group ($t_{cluster} = 64.63$, $t_{max} = 7.31$, $p < .001$), and from −1000 to 2000 ms in the 36-month-old group ($t_{cluster} = 191.50$, $t_{max} = 10.63$, $p < .001$).

Individual differences

Analyses of individual differences in anticipatory effects showed that none of the models including measures of vocabulary, word combinations, syntactic complexity, or cognitive development significantly improved model fit compared to the reference models including only Condition (all p-values > .29). This pattern was consistent across all age groups (24, 30, and 36 months). Because these exploratory analyses may have been limited by statistical power and age-specific measurement differences, we do not draw strong conclusions from these null effects (Table 4).

Experiment 1 Discussion

Across groups, children rapidly shifted their gaze toward the Target once the noun was heard (integration window), confirming robust bottom-up recognition. Crucially, anticipatory looking before noun onset showed an age-dependent pattern: the Bonferroni-adjusted Gender-matched > Unrelated follow-up contrast—our key index of gender-based prediction—did not reach significance at

Table 4
Individual-Difference Exploration for Experiment 1

Group	Fixed structure	N	AIC	LogLik	χ^2	p
24	Condition * Vocabulary	14	2590.700	−1281.3	2.560	0.465
24	Condition * Word Combinations	14	2592.200	−1282.1	1.017	0.797
24	Condition * Syntactic Complexity	14	2591.800	−1281.9	1.425	0.700
30	Condition * Vocabulary	14	2530.700	−1251.4	3.748	0.290
30	Condition * Word Combinations	14	2533.300	−1252.7	3.146	0.766
30	Condition * Syntactic Complexity	14	2531.6	−1251.8	2.9051	0.406
30	Condition * Cognitive development	14	2533.600	−1252.8	1.222	0.748
36	Condition * Cognitive development	14	1651.200	−811.60	1.023	0.796

Note. All models were compared directly against the reference model, including only Condition as a fixed effect.The dependent variable was the empirical logits. Random structure: (Condition|Subject) + (1|Item).AIC: Akaike Information Criterion; LogLik: log-likelihood.N = number of parameters.All model comparisons have 3 degrees of freedom.The Bonferroni-corrected significance threshold was α = .00625.

24 months but was statistically reliable within the 30- and 36-month groups. The significant Gender-matched × 36 months interaction indicated that this effect was larger at 36 months than at 24 months, whereas the nonsignificant Gender-matched × 30 months interaction provided no evidence that the effect differed between 30 and 24 months. The cluster-based time-course analyses provided a descriptive complement to the window-based results: in the 36-month group, the significant Gender-matched > Unrelated cluster overlapped with the final 1000 ms of the silent pre-noun interval and continued after noun onset, whereas the clusters identified in the younger groups were confined to the post-noun interval. However, because cluster-based permutation tests do not provide precise inferential estimates of effect onset or offset, and the cluster boundaries were not statistically compared across age groups, these apparent temporal differences should be interpreted descriptively rather than as evidence of significant differences in latency or duration (Sassenhagen & Draschkow, 2019).

Because possessive determiners in our design (e.g., su/sus "their," mi "my") do not encode grammatical gender in Spanish, children could not rely on agreement with a preceding gender-marked determiner. Anticipatory looks to the Gender-matched object therefore suggest that children use contextual information to generate expectations about the upcoming noun and its associated gender feature. The observed developmental pattern indicates that the use of morphosyntactic information becomes more robust across the third year of life, while younger children's behavior remains more closely tied to bottom-up confirmation following noun onset. The individual-difference analyses did not provide clear evidence that vocabulary, word combinations, syntactic complexity, or cognitive scores explained additional variance in anticipatory looking. However, because these analyses were exploratory and may have been limited by statistical power and age-specific measurement differences, we do not draw strong conclusions from these null effects.

However, nouns in this experiment had regular, prototypical endings (–a, typically marking feminine nouns, –o, typically marking masculine nouns), raising the possibility that anticipatory behavior may have been driven by sensitivity to surface phonological regularities rather than by the pre-activation of abstract gender features.

Crucially, anticipatory looks toward the Gender-matched object require that participants not only activate expectations about the upcoming target noun, but also pre-activate its associated grammatical gender feature, which in turn guides visual attention toward items sharing that feature. However, in the present design, it remains possible that children relied on expectations about phonological form (e.g., nouns ending in –a or –o), rather than on grammatical gender per se, to guide anticipatory looking.

Experiment 2 was therefore designed to address this limitation by removing transparent gender-ending cues from the Gender-matched objects. By using nouns that did not provide prototypical –o/–a gender-ending cues, this experiment allowed us to test whether anticipatory effects were driven by the pre-activation of lexical gender representations rather than by surface-level phonological cues.

Experiment 2: Gender Prediction Without Transparent Gender-Ending Cues

Method

Participants

The experiment included three age groups: 48 Mexican Spanish-learning children at 24 months (M = 24.03 months, SD = 0.28; 25 boys), 48 children at 30 months (M = 30.01 months, SD = 0.32; 26 boys), and 48 children at 36 months (M = 36.01 months, SD = 0.28; 28 boys). These sample sizes were selected to maintain statistical sensitivity comparable to that established in Experiment 1 while also accommodating the level of trial loss typically expected in eye-tracking research with young children. All participants were

Table 5
Participant assessment descriptives by age group

Domain	Measure	24 months M (SD), range	30 months M (SD), range	36 months M (SD), range
Socioeconomic background	Monthly household income	MXN 12,219 (1,001), 10,396–13,973	MXN 12,260 (1,114), 10,443–13,938	MXN 11,746 (1,067), 10,223–13,973
CDI-II	Words produced	498.56 (29.09), 454–554	490.23 (24.33), 450–534	—
	Word combinations	6.04 (0.82), 5–7	5.33 (1.08), 4–7	
	Syntactic complexity	13.12 (2.29), 9–16	26.60 (2.74), 21–32	
WPPSI-III	Puzzles	—	16.73 (1.09), 15–18	19.25 (2.88), 15–24
	Block Design		22.35 (2.14), 19–25	24.48 (3.14), 20–29
	Information		15.21 (1.88), 12–18	17.58 (2.25), 14–21
	Receptive Vocabulary		20.88 (2.04), 18–24	22.21 (2.49), 18–26
	Naming		15.23 (1.12), 14–17	18.79 (2.15), 16–22
	Performance IQ	—	101.81 (4.54), 91–109	97.75 (9.23), 82–115
	Verbal IQ	—	100.38 (6.60), 85–115	101.06 (8.85), 85–121
	Full-scale IQ	—	101.02 (4.04), 92–108	99.58 (6.50), 87–112

Note. Values are M (SD), range, unless otherwise indicated. Monthly household income is reported in Mexican pesos. As a national benchmark, the 2024 average monthly household income in Mexico was MXN 25,955 according to the National Survey of Household Income and Expenditure (ENIGH; INEGI, 2025); therefore, the sample means reported here were below the national household average and should be interpreted as descriptive socioeconomic information. CDI-II = MacArthur–Bates Communicative Development Inventories. CDI-II words produced corresponds to the number of words children were reported to produce; word combinations indexes the number of words children combined in utterances; syntactic complexity indexes children's use of more complex syntactic structures. WPPSI-III = Wechsler Preschool and Primary Scale of Intelligence, Third Edition. Dashes indicate that the measure was not administered to that age group.

monolingual Spanish speakers with normal hearing and vision and no reported neurological or language impairments, based on parental reports. Participant socioeconomic information and standardized assessment descriptives, including ranges, are summarized in Table 5.

Experimental design and stimuli

The experimental design was identical to that of Experiment 1; the same temporal structure, counterbalancing scheme, and three display conditions were used: Target, Gender-matched, and Unrelated. Thus, each participant saw each sentence only once, in one condition, and completed 21 trials in total: 7 Target, 7 Gender-matched, and 7 Unrelated trials.

The sentences used in Experiment 2 were the same as those in Experiment 1. However, the critical difference concerned the gender-matched object. In Experiment 1, the gender-matched objects shared the grammatical gender of the expected target noun and carried transparent, prototypical gender endings in Spanish, such as –o for masculine nouns and –a for feminine nouns. In Experiment 2, these objects were replaced with gender-matched nouns that shared the target noun's grammatical gender but did not provide transparent –o/–a gender-ending cues. This manipulation provided a stricter test of whether children could pre-activate grammatical gender when neither the possessive determiner nor transparent noun-ending cues provided overt gender information before noun onset.

For example, for the sentence frame *La gallina puso su… huevo* "The hen laid its… egg," the expected noun was *huevo* "egg," a masculine noun. In the Gender-matched condition, the target picture was absent, and the gender-matched object was *reloj* "clock," also masculine but lacking the prototypical masculine –o ending. As in Experiment 1, the gender-matched objects and distractors were selected to be semantically and phonologically unrelated to the expected target noun and to the sentence context.

Full stimulus lists, counterbalancing information, phonological and semantic similarity checks, image-naming results, and visual-saliency analyses are reported in the Supplementary Materials.

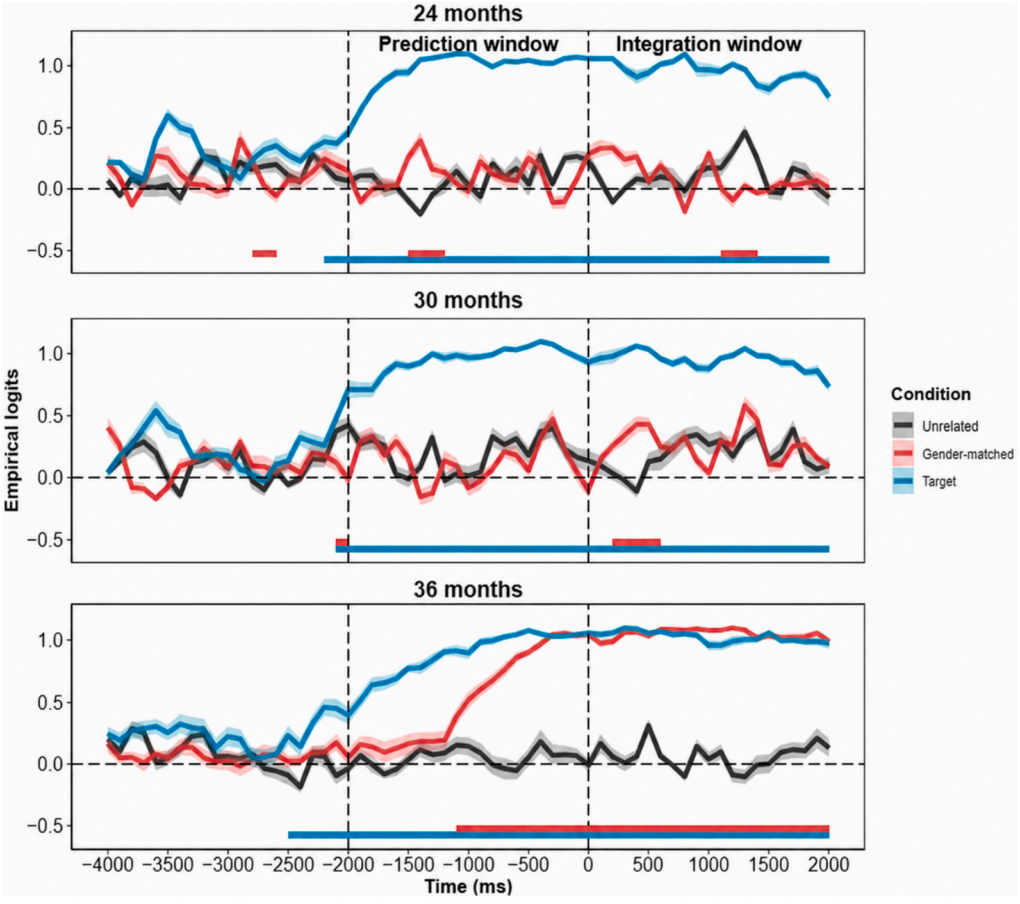


Fig. 3. Fixation Time Course for Experiment 2. Note. Lines represent the mean empirical logits, with shaded ribbons indicating ± 1 standard error. Vertical dashed lines mark the analysis windows, and horizontal dashed lines indicate chance level. Solid horizontal bars at the bottom denote significant clusters: red for the Gender-matched–Unrelated contrast, and blue for the Target–Unrelated contrast.

Data Processing and Statistical Analysis

The data processing and statistical analysis followed the same procedures as those used in Experiment 1. All participants contributed at least one valid trial per condition, with an overall average of 19.75 valid trials (*SD* = 2.33, *range* = 10 - 21) out of 21. In the 24-month-old group, participants contributed an average of 6.83 target trials (*SD* = 0.37, *range* = 6 - 7), 6.75 Gender-matched trials (*SD* = 0.60, *range* = 5 - 7), and 7.00 unrelated trials (*SD* = 0.00, *range* = 7 - 7). In the 30-month-old group, the averages were 6.75 (*SD* = 0.43, *range* = 6 - 7), 6.75 (*SD* = 0.43, *range* = 6 - 7), and 7.00 (SD = 0.00, *range* = 7 - 7) for target, Gender-matched, and unrelated trials, respectively. For the 36-month-old group, participants contributed an average of 6.00 target trials (*SD* = 1.28, *range* = 3 - 7), 6.00 Gender-matched trials (SD = 1.12, *range* = 4 - 7), and 6.18 unrelated trials (SD = 1.43, *range* = 2 - 7).

Results

Linear mixed effect models

Fig. 3 illustrates the time course of fixation patterns across the analysis windows, and Table 6 presents the full statistical values of the linear mixed-effect models. In the prediction window, only the fixed effect of Target was significant ($p < 0.001$), indicating that the average empirical logit of fixations was significantly higher in the Target condition than in the Unrelated condition at 24 months. The Group effect for 30 months was also significant ($p = 0.048$), suggesting a higher overall fixation rate at 30 months compared to 24 months. The Gender-matched × 36 months interaction was also significant ($p < 0.001$), indicating a significantly stronger gender-matched effect at 36 months relative to 24 months. Other interactions were not significant.

Bonferroni-adjusted follow-up contrasts showed that in the 24-month-old group, there was no significant difference in fixations between the Unrelated and Gender-matched conditions ($\beta = 0.005$, $SE = 0.048$, $df = 136$, $t = 0.101$, $p > 0.999$), indicating similar fixation rates across the two conditions. Similarly, in the 30-month-old group, fixation proportions again did not significantly differ between the Unrelated and Gender-matched conditions ($\beta = 0.074$, $SE = 0.048$, $df = 136$, $t = 1.533$, $p = 0.383$). Finally, in the 36-month-old group, fixations were significantly higher in the Gender-matched than in the Unrelated condition ($\beta = -0.582$, $SE = 0.051$, $df = 160$, $t = -11.424$, $p < 0.0001$).

In the integration window model, the fixed effects of Target ($p < 0.001$) and 30 months ($p = 0.027$) were significant, indicating that fixations were significantly higher in the Target condition than in the Unrelated condition at 24 months, and that overall fixation levels were higher at 30 months than at 24 months. Additionally, the Gender-matched × 36 months interaction was significant ($p < 0.001$), indicating a larger gender-matched effect at 36 months compared to 24 months. No other interaction terms were significant.

Bonferroni-adjusted follow-up contrasts showed that, in the 24-month-old group, fixations did not significantly differ between the Unrelated and Gender-matched conditions ($\beta = -0.012$, $SE = 0.046$, $df = 134$, $t = -0.264$, $p > 0.999$), indicating similar fixation levels across the two conditions.

In the 30-month-old group, the difference between the Unrelated and Gender-matched conditions was not significant ($\beta = -0.053$, $SE = 0.046$, $df = 137$, $t = -1.154$, $p = 0.751$), indicating comparable fixation levels across the two conditions.

In the 36-month-old group, fixations were significantly higher in the Gender-matched than in the Unrelated condition ($\beta = -1.088$, $SE = 0.048$, $df = 158$, $t = -22.582$, $p < 0.001$).

To verify that the Experiment 2 findings were not driven by the inclusion of two items that were irregular rather than opaque (mano "hand" and moto "motorbike," both feminine nouns ending in –o), we conducted a sensitivity analysis excluding all trials containing these items. The prediction- and integration-window mixed-effects models were refitted using the same fixed- and random-effects structures as in the primary analyses. This exclusion did not change the pattern of results: the gender-matched-object advantage remained absent at 24 and 30 months but was robust at 36 months in both analysis windows, and the target advantage remained reliable. Thus, the main findings of Experiment 2 were not driven by these two items. Full model estimates and follow-up contrasts are reported in the Supplementary Materials.

Table 6
Linear Mixed-Effects Models for Experiment 2

Fixed effects	Prediction window					Integration Window				
	β	*SE*	*df*	*t*	*p*	*β*	*SE*	*df*	*t*	*p*
Intercept	-0.053	0.038	118.808	-1.402	0.163	**-0.084**	**0.04**	**95.41**	**-2.085**	**0.039**
Gender-matched	-0.004	0.047	184.168	-0.102	0.919	0.012	0.045	180.4	0.266	0.79
Target	**1.046**	**0.046**	**932.248**	**22.352**	**< 0.001**	**1.042**	**0.043**	**437.2**	**23.957**	**< 0.001**
30 months	**0.093**	**0.047**	**177.825**	**1.99**	**0.048**	**0.104**	**0.047**	**129.1**	**2.228**	**0.027**
36 months	-0.049	0.048	191.488	-1.011	0.313	0.02	0.048	136.6	0.415	0.678
Gender-matched : 30 months	-0.068	0.067	183.974	-1.023	0.307	0.04	0.063	181.9	0.638	0.524
Target : 30 months	-0.128	0.066	940.451	-1.939	0.052	-0.102	0.061	445.1	-1.665	0.096
Gender-matched : 36 months	**0.587**	**0.069**	**199.54**	**8.464**	**< 0.001**	**1.076**	**0.065**	**195.5**	**16.382**	**< 0.001**
Target : 36 months	-0.051	0.068	987.751	-0.758	0.448	0.048	0.063	474	0.762	0.446

Note. Bold values indicate statistical significance.

Cluster-based permutation analysis

In the Target vs. Unrelated contrast, toddlers generally looked more toward the Target than toward the Unrelated object. In the 24-month-old group, we found a large cluster extending from −2200 to 2000 ms (tcluster = 555.33, tmax = 26.55, p < .001). In the 30-month-old group, a significant cluster also spanned the −2000 to 2000 ms window (tcluster = 434.91, tmax = 19.52, p < .001). Similarly, in the 36-month-old group, a significant effect was found from −2500 to 2000 ms (tcluster = 542.57, tmax = 19.62, p < .001).

By contrast, the Gender-matched vs. Unrelated comparisons revealed a more variable pattern across age groups. In the 24-month-old group, three significant clusters were observed. In the first cluster, from −2800 to −2600 ms, the Unrelated condition elicited more looks than the Gender-matched condition ($t_{cluster} = -10.39$, $t_{max} = -2.32$, $p = .019$). In the second cluster, from −1500 to −1200 ms, the Gender-matched condition elicited more looks ($t_{cluster} = 12.71$, $t_{max} = 4.53$, $p = .001$). The third cluster, from 1100 to 1400 ms, again showed increased fixations toward the Unrelated condition ($t_{cluster} = -11.74$, $t_{max} = -2.34$, $p = .003$). In the 30-month-old group, a significant cluster from −2100 to −600 ms indicated increased fixations toward the Unrelated condition ($t_{cluster} = -10.40$, $t_{max} = -4.44$, $p = .013$), while a second cluster from 300 to 700 ms showed more looks to the Gender-matched condition ($t_{cluster} = 18.55$, $t_{max} = 4.95$, $p < .001$). Finally, in the 36-month-old group, a robust cluster from −1100 to 2000 ms indicated increased fixations toward the Gender-matched condition compared to the Unrelated condition ($t_{cluster} = 432.20$, $t_{max} = 20.71$, $p < .001$).

Individual differences

Models including word combinations, syntactic complexity, or cognitive development did not improve model fit compared to the reference models including only Condition (all p-values > .052). This pattern was consistent across all age groups (Table 7). Vocabulary showed a nominally significant model-comparison effect at 30 months (p = .015), but this result did not meet the Bonferroni-corrected significance threshold of α = .00625. We therefore treat this age-specific effect as exploratory and do not interpret it as evidence for a robust relation between vocabulary and gender-based prediction.

Experiment 2 Discussion

When transparent gender-ending cues were removed from the gender-matched objects, toddlers across age groups showed robust Target > Unrelated looking both before and after noun onset. This pattern indicates that children were able to use contextual information to anticipate the upcoming noun when the target picture was present, even though the possessive frames did not provide an overt pre-nominal gender cue. Thus, the Target effect confirms context-based lexical anticipation, but it does not by itself show that children pre-activated grammatical gender.

In contrast, the gender-matched-object effect—our indicator that children pre-activated gender information associated with the expected noun—was not reliable at 24 or 30 months and was reliably observed only at 36 months. This pattern was observed in both the prediction and integration windows. The cluster-based analyses provided a complementary descriptive view of the time course: the younger groups showed brief and inconsistent Gender-matched versus Unrelated differences, whereas the 36-month group showed a significant Gender-matched > Unrelated cluster spanning part of the silent pre-noun interval and the post-noun period. However, because cluster-based permutation tests do not provide precise inferential estimates of effect onset or offset, and the cluster boundaries were not statistically compared across age groups, these apparent temporal differences should be interpreted descriptively rather than as evidence of significant differences in latency or duration (Sassenhagen & Draschkow, 2019). In Experiment 1, by comparison, the within-group Gender-matched > Unrelated contrast was already statistically reliable at 30 months when transparent gender-ending cues were available, although no direct statistical comparison of effect magnitude across experiments was conducted. Together, these results suggest that younger children can anticipate upcoming lexical items under strong contextual constraint, but reliable anticipatory access to grammatical gender becomes evident later when children cannot rely on transparent surface-form cues.

This dissociation indicates that anticipatory processing at the lexical level does not necessarily entail access to morphosyntactic features. Instead, when transparent –o/–a gender-ending cues are unavailable in the gender-matched object, mapping contextual

Table 7
Individual-Difference Exploration for Experiment 2

Group	Fixed structure	N	AIC	LogLik	χ^2	p
24	Condition * Vocabulary	14	2050	−1010.8	0.634	0.889
24	Condition * Word Combinations	14	2049	−1010.5	1.089	0.78
24	Condition × Syntactic Complexity	14	2049	−1010.6	0.929	0.818
30	**Condition * Vocabulary**	**14**	**2031**	**−1001.4**	**10.49**	**0.015**
30	Condition * Word Combinations	14	2040	−1005.8	1.804	0.614
30	Condition × Syntactic Complexity	14	2037	−1002.8	7.717	0.052
30	Condition * Cognitive development	14	2040	−1006.1	1.109	0.775
36	Condition * Cognitive development	14	1634	−802.8	2.71	0.439

Note. All models were compared directly with the reference model, including only Condition as a fixed effect.The dependent variable was the empirical logits. Random structure: (Condition|Subject) + (1|Item).AIC: Akaike Information Criterion; LogLik: log-likelihood. N = number of parameters. All model comparisons have 3 degrees of freedom. The Bonferroni-corrected significance threshold was α = .00625.

information onto grammatical gender appears to place additional demands on the system, which were consistently met only by 36 months. In this sense, Experiment 2 provides a stricter test of gender-based prediction because anticipatory looking to the gender-matched object could not be explained by simple matching of prototypical noun endings. Rather, the effect requires children to use the sentence context to activate the expected noun and retrieve gender information associated with that noun's lexical representation.

One feature of the time-course results deserves comment. In the Target > Unrelated contrast, significant clusters included portions of the end of the sentence context and the transition into the silent interval. Descriptively, this pattern is consistent with children gradually accumulating contextual constraint as the sentence unfolded. This early target-related looking is consistent with lexical anticipation. However, the critical evidence for gender-based prediction comes from the Gender-matched > Unrelated contrast, because the expected target picture was absent in that condition. At 36 months, the significant Gender-matched > Unrelated cluster overlapped with the silent interval, during which no additional auditory information was available. This overlap is consistent with the interpretation that older children used internally generated expectations to pre-activate gender information associated with the expected noun. However, the precise onset of the effect cannot be inferred from the observed cluster boundary.

Age differences in overall fixation levels, with higher fixation values at 30 than at 24 months, should be interpreted cautiously. Although they may reflect developmental improvements in sustained attention or task engagement, they do not constitute evidence for stronger gender-based prediction, given the absence of a reliable pre-noun gender-matched-object advantage before 36 months. Similarly, the individual-difference analyses were exploratory. Although productive vocabulary showed a nominal association with looking patterns at 30 months, this effect did not survive the Bonferroni-corrected threshold and concerned the Target–Unrelated contrast rather than the critical Gender-matched–Unrelated contrast. We therefore treat this finding as tentative and do not interpret it as evidence that vocabulary supported gender-based prediction. Likewise, we do not draw strong conclusions from the absence of reliable effects in the other individual-difference models.

Together with Experiment 1, these findings support a graded developmental trajectory: children can use strong sentence contexts to anticipate upcoming lexical items early in the third year, but anticipatory access to grammatical gender in the absence of transparent form cues becomes reliable later. Experiment 2 therefore strengthens the conclusion that the gender-matched-object advantage observed at 36 months reflects access to lexical–morphosyntactic representations, rather than reliance on superficial form-based cues such as prototypical –o/–a endings.

General Discussion

Across two visual-world eye-tracking experiments, the present study examined how children's use of sentence context to guide real-time anticipation changes between 24 and 36 months of age. By introducing a silent interval before the final noun, the design allowed us to separate anticipatory looking from later word recognition. Across both experiments, children showed evidence that highly constraining sentence contexts can guide attention before the expected word is heard. However, the specificity of this anticipation changed with age. Whereas younger children showed evidence of anticipating the expected target under strong contextual constraint, looks to the Gender-matched object followed a more selective developmental pattern. When transparent gender-ending cues were available, the window-based Gender-matched > Unrelated contrast was statistically reliable within both the 30- and 36-month groups. The significant Gender-matched × 36 months interaction showed that this effect was larger at 36 months than at 24 months, whereas the nonsignificant Gender-matched × 30 months interaction provided no evidence that the effect differed between 30 and 24 months. No direct comparison between the 30- and 36-month groups was conducted. In the complementary time-course descriptive analysis, a significant Gender-matched > Unrelated cluster overlapped with the silent pre-noun interval at 36 months, whereas the clusters observed in the younger groups were confined to the post-noun period. When transparent gender-ending cues were removed, reliable Gender-matched-object anticipation was observed only at 36 months.

Placed alongside prior evidence that semantic prediction emerges around 18–24 months (Angulo-Chavira et al., 2025; Mani & Huettig, 2012) and phonological prediction near 30 months (Angulo-Chavira et al., 2025), the present results support a graded developmental trajectory: meaning pre-activation consolidates first; pre-activation of form follows; and only later do abstract morphosyntactic features such as grammatical gender become available for real-time pre-activation when no overt functional marker is present. Importantly, this developmental ordering should not be taken to mean that young children cannot use gender information earlier in comprehension. Prior studies have shown that infants and toddlers can use overt gender-marked determiners or transparent gender-related cues to facilitate word recognition and anticipation (e.g., Lew-Williams & Fernald, 2007; Mornati et al., 2023; van Heugten & Johnson, 2011). The present study addresses a different question: whether children can generate gender-based expectations when the pre-nominal speech signal does not itself provide overt gender information.

Theoretically, our findings are best interpreted within multi-component accounts of predictive processing that allow contributions from associative/semantic, combinatorial/statistical, and production-related pathways rather than committing to a single mechanism (Huettig, 2015; Kuperberg & Jaeger, 2016; Pickering & Gambi, 2018). Prediction-by-association accounts propose that predictions arise from learned links between contextual cues and lexical representations, whereas prediction-by-production accounts posit that comprehenders internally simulate production processes to generate expectations about upcoming words and their features. In our data, children anticipated upcoming nouns before noun onset, indicating that contextual information was sufficient to activate lexical candidates. The key developmental change concerned whether this lexical anticipation also activated grammatical gender information strongly enough to guide looks toward a semantically and phonologically unrelated object that shared the expected noun's gender. Thus, the data are compatible with lexicon-mediated top-down activation, in that predictions appear to depend on access to stored lexical representations. In Experiment 1, where Gender-matched objects carried prototypical gender-marking endings, the Gender-matched-object advantage observed within the 30-month group may reflect children's developing sensitivity to distributional

regularities linking phonological form and grammatical gender. However, the reliable anticipatory effect observed at 36 months in Experiment 2—where transparent gender-ending cues were removed—suggests that prediction cannot be fully explained by reliance on phonological cues alone and instead reflects access to lexicalized gender representations.

While lexicon-mediated top-down pre-activation is compatible with a prediction-by-production mechanism (Pickering & Gambi, 2018), the anticipatory effects reported here can be explained without assuming full production-based simulation of upcoming forms. Specifically, Experiment 1's findings are consistent with only partial pre-activation of upcoming forms (i.e., of the regular noun endings), which could be mediated either by production-based simulation mechanisms or by associative mechanisms. In contrast, Experiment 2's findings provide evidence for pre-activation of a semantically predictable noun's gender when transparent surface-form cues were unavailable. However, the present analyses do not allow us to establish temporal precedence between effects (i.e., whether gender pre-activation temporally followed meaning pre-activation, as would be expected under production-based prediction mechanisms). More generally, the results align with accounts in which bottom-up constraints and cue availability modulate the extent and content of prediction early in development (Chow et al., 2022): when reliable surface cues are absent, predictive processing places greater demands on lexical access and representational stability.

The cluster-based analyses provided a complementary, descriptive view of the temporal distribution of these effects. For the Target > Unrelated contrast, significant clusters overlapped with the end of the sentence context and/or the transition into the silent interval in some groups. This pattern is consistent with children gradually accumulating contextual constraint as the sentence unfolded, rather than responding only after the spoken target noun. The critical evidence for gender-based anticipation, however, comes from the Gender-matched > Unrelated contrast. In Experiment 2, the significant cluster observed in the 36-month group overlapped with the silent pre-noun interval and continued after noun onset, a pattern consistent with internally generated expectations activating gender information associated with the expected noun. However, cluster-based permutation tests support inference at the level of the cluster rather than precise estimates of effect onset or offset, and the observed cluster boundaries were not statistically compared across conditions or age groups. Accordingly, apparent differences in effect latency or duration should be interpreted descriptively and not as statistically established timing differences (Sassenhagen & Draschkow, 2019).

Relative to studies in which gender information is overtly provided by determiners or other pre-nominal elements, the present design imposed additional representational demands because children had to generate the relevant gender expectation without hearing a gender-marked cue. For this reason, the later emergence of robust gender-based anticipation in the present study should be interpreted as specific to cue-free, internally generated gender prediction, rather than as evidence that children generally lack access to grammatical gender before 36 months. Indeed, prior work shows earlier use of gender information when overt cues are available, including evidence from infants and younger toddlers (e.g., Mornati et al., 2023; van Heugten & Johnson, 2011).

Regarding individual differences, the analyses involving CDI-II and WPPSI-III measures were exploratory and should be interpreted cautiously. Overall, adding individual-difference predictors did not consistently improve model fit. Although a nominal vocabulary-related association appeared at 30 months in Experiment 2, this effect did not survive the Bonferroni-corrected threshold and concerned the Target–Unrelated contrast rather than the critical Gender-matched–Unrelated contrast. We therefore do not interpret this association as evidence for a robust role of individual differences in gender prediction. Likewise, the absence of reliable effects should not be taken as evidence that vocabulary, syntax, or cognitive skills are unrelated to predictive processing; these exploratory analyses may have had limited sensitivity because predictors were tested separately within age groups and the assessment batteries differed across cohorts.

These findings suggest two directions for future research. First, because our sentences were highly restrictive, future work should parametrically vary contextual constraint to estimate the level of contextual support required for gender to become predictive earlier in development. Second, the typological scope should be broadened to include languages with less transparent gender systems (e.g., noun-class languages such as Swahili or Zulu), testing whether the sequencing and timing observed here generalize.

Finally, to refine mechanistic inference, future studies should manipulate lexical frequency and gender consistency in the input. Increasing the frequency with which specific nouns appear in predictive contexts would strengthen associations between sentence context and lexical items, whereas manipulating gender consistency would directly test the extent to which children rely on regularities linking form and gender.

This design would allow dissociating the contributions from statistical learning (prediction-by-association) and production-based mechanisms (prediction-by-production), as well as probing potential interactions among semantic, phonological, and morphological pathways (Dell & Chang, 2014; Pickering & Gambi, 2018). Combining these manipulations with independent measures of children's lexical and cognitive development (e.g., vocabulary size or processing speed) would help characterize the dynamics underlying morphological anticipation without presupposing a single predictive mechanism.

Taken together, the findings show that children's ability to anticipate upcoming spoken language becomes increasingly precise between 24 and 36 months. Strong sentence context was sufficient to guide anticipatory looks to expected targets across age groups, but gender-matched-object anticipation showed a more restricted developmental profile. This dissociation suggests that predicting a likely word and using that prediction to activate more specific properties of the word are separable achievements in early comprehension. More broadly, the study provides experimental evidence that real-time language prediction in early childhood is constrained by the stability and accessibility of children's lexical representations. By 36 months, children can use context not only to anticipate what word is likely to occur, but also to activate grammatical gender information associated with that word before it is heard, even when overt pre-nominal gender cues and transparent gender-ending cues are unavailable.

Conclusions

Taken together, these findings yield three main contributions. First, they provide evidence that grammatical gender becomes available for anticipatory processing during the third year of life, following earlier-emerging semantic prediction. Second, they show that gender-based anticipation does not depend solely on prototypical –o/–a endings but instead relies on lexical knowledge of gender that can be activated by contextual information. Third, they introduce a sensitive methodological framework to examine the temporal dynamics of predictive processing during early language development, which can be extended to other morphosyntactic features, such as number or person, and to typological contrasts across languages with different gender systems.

More broadly, the present findings are consistent with the view that different sources of information contribute to prediction at different stages of development, with earlier reliance on semantic constraints and later incorporation of morphosyntactic features as representations become more stable and accessible. This characterization provides a developmentally grounded account of how predictive processing evolves in early language comprehension while leaving open the precise mechanisms through which these effects arise. Future work should determine how associative learning, production-based mechanisms, lexical frequency, and cross-linguistic differences in gender marking jointly shape the emergence of morphosyntactic prediction.

CRediT authorship contribution statement

Alejandra Mitzi Castellón-Flores: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Armando Quetzalcóatl Angulo-Chavira:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Chiara Gambi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Natalia Arias-Trejo:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The eye-tracking data and R analysis scripts used for Experiment 1 and Experiment 2 are openly available on the Open Science Framework (OSF): https://osf.io/xzgvk/overview?view_only=22d71a4173ed4048bef52b63d800cb45.

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